

		Using trigonometry to deduce the angle.
5	D	The amount of weight that cause the extension of each spring is $\frac{1}{3}W$. $\Rightarrow \frac{1}{3}W = kx \Rightarrow k = \frac{W}{3x}$ When weight of $2W$ is hung from the beam supported by two identical springs, amount of weight that cause the extension of each spring is $\frac{1}{2}W$. $\Rightarrow W = \frac{W}{3x}x' \Rightarrow x' = 3x$
6	B	Work done in stretching the material from 0 to T is area under